

Investigation of failure mechanism of stator winding and stator core. Protocol for repair and acceptance testing of stator winding and stator core. Asset management strategy for stator assembly for machines above 100MVA.

Yuriy Kharchenko, P.Eng.
Kemano G.S., Rio Tinto
Kitimat, British Columbia
Yuriy.Kharchenko@riotinto.com

Inna Kremza P.Eng.
Voith Hydro
Mississauga, Ontario
Inna.Kremza@voith.com

Abstract— A failure of the stator winding and core happens more often in service than it can be predicted during offline and online diagnostics. Failures in stator assembly can be caused by various factors mainly affecting its insulation system, including electrical, mechanical, and thermal stresses. Online stator failures machines rated above 100 MVA, may result in significant equipment downtime and repair costs. This paper outlines the methodology for effective investigation, repair and testing of stator failures to minimize the economic losses and extend useful life of the equipment following the failure. Moreover, asset management strategy is described to assist equipment owner/operator with full lifecycle management of the stator assembly.

Keywords— *failure mechanisms; stator winding; core; repair; acceptance testing; asset management; electric machines, above 100MVA.*

I. INTRODUCTION

Power plants are required to efficiently meet the electrical demand, and when feeding metal smelters unit load and utilization factor can reach 95%. Appropriate and swift generator failure analyses and repair protocols are essential for ensuring the plant reliability and availability. This paper proposes the framework for effective investigation, repair and testing of stator failures illustrated by a case study of hydro generator. In many cases of failure, the owner have the option to either rewind the generator or proceed with repairs if feasible. A new winding is expected to have a service life of more than 35 years, and repairs can extend the life of the stator winding, providing time to prepare for a full rewind and comprehensive evaluation of the machine.

This case study focuses on a hydro generator rated 132 MVA, 13.8 kV, and 327 rpm synchronous speed powered by Pelton turbine as a prime mover. The generator was constructed in 1956 with Roebel bar wound with two parallel circuits per phase and generator neutral grounding transformer. In 1986 this generator was rewound and the original four (4) splits core was reused. After 32 years in service, the unit experienced a ground failure of the stator winding. The proposed data-driven approach

is applied to the case study of a hydroelectric plant, utilizing real-life data.

II. GENERATOR FAILURE INVESTIGATION

The investigation of failures within a hydropower plant using IEEE standards [1, 2] involves a comprehensive analysis of the generator's electrical and mechanical components. The failure mode analysis begins with protection data review, a visual inspection of the unit that involves a thorough examination of all assembly components (the stator core, windings, connections, support ring, wedging system, etc.) and followed by electrical testing to identify faulty subcomponents.

A. Generator Protection Targets Review

The generator protection included two redundant generator protection relays. The protection scheme detected fault in operation and tripped the unit via activating elements 59GX1 and 64G [3], indicating a ground fault on C-phase.

B. Stator Winding Visual Inspection

After unit disassembly and rotor removal a visual inspection of the stator winding revealed a fault zone in the area of two slots. A significant amount of oil was found under the slot wedges, inside the slot, and in the insulating caps. The top bar in one of the slots showed traces of electrical discharges due to the failure, but no visible damage was detected in the rest of the winding. The terminals and support structure of the winding, such as support rings and parallel rings, showed normal aging conditions with some soft or dry spots due to the degradation of the composite material/epoxy. The tapping test of the wedges detected a few loose wedges around the bore but not of significant concern.

C. Stator Winding Electrical Testing

Insulation resistance IR testing and polarization index measurement PI were completed as per IEEE practices [4] at voltage 5 kV(for 13.8 kV machine) for all six (6) circuits. The fault was confirmed on C-phase. The insulation resistance readings and polarization factor for uncompromised circuits were IR>10 G ohms and PI in the range between 5-6. (Fig. 1)

High voltage DC ramp test was completed as per IEEE standards [2, 5] at voltage level of 23.46 kV DC (i.e. 1.7U ac nom was selected as DC equivalent voltage) and ramp rate 1 kV/min for all healthy circuits and excluding faulty one. Normal leakage current was observed for all tested circuits in the range of 5-6 uA/circuit with very small absorption current component.

Based on winding diagram it was identified that both top and bottom bars of the slot were part of the same circuit. In order to identify the faulty bar the top bar was unbrazed and removed from the slot (Fig.4). The insulation resistance test was repeated for two segments of the open circuit with no fault confirmed in either segment. HVDC ramp test was conducted on faulty circuit to ensure HV withstand of remaining bars ground wall insulation yield in positive results.

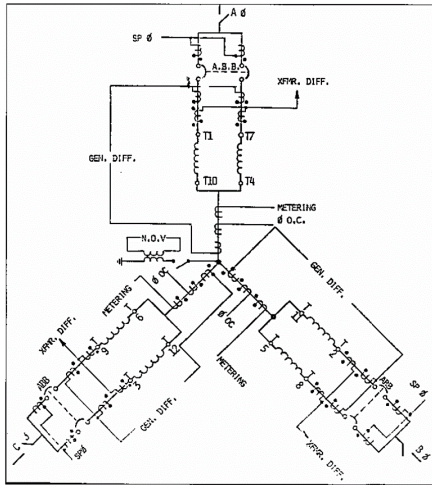


Fig. 1 Generator Winding Schematics Two Circuits per Phase

D. Stator Core Visual Inspection and Knife Test

After rotor removal the damage to the core was easily identified and hence a failure mode of the generator (Fig. 2). The support finger for lower core tooth packet between slots was displaced downwards and about 45% of this tooth packet laminations were destroyed by the extreme temperature and erosion. It was apparent that as soon a compression of the tooth packet laminations was decreased and the insulating layer on the laminations wore out due to vibration allowing the development of eddy current and subsequently creating hot spot in the area. The hot spot in this core tooth packet eventually led to localized core damage and bar failure in adjacent to packet slot. Migration of the laminations from burned tooth packet could also contribute to the ground wall insulation failure due to additional mechanical pressure, however inspection of side packing and bar dissection did not revealed that evidence. Bar dissection confirmed temperature and electric nature of the ground wall insulation fault due to dry and delaminated state of the insulation as well as core hot spot as a main cause.

A visual inspection [1] of the remaining stator core revealed signs of aging after being in use for 65 years. The core exhibited waving and chevroning at the splits. Ventilation spacings were reduced throughout the entire stack due to the accumulation of dirt and flaring at the edges of the bore. To inspect the tight spaces, a flexible borescope was used to inspect the vent spacing

in the stator core and the condition of the stator winding in the slot.



Fig. 2 Core Damage Area

The knife insertion test was conducted at the back of the stator core for evaluation of the overall core compression. The test revealed loose lamination at the bottom of the stator core and the burned tooth packet area, with knife penetration reaching a depth of 1.2 inches.

E. Stator Core EL CID

The core surface was cleaned, and electromagnetic core imperfection detection test (EL CID) was conducted to identify any other area of the core that may need a repair.

The industry standard practice conducting EL CID test with a flux density equal to 4% of nominal density [6] to achieve the magnetic flux response for a healthy core at level below of 100 mA with a Chattock coil sensor placed on the core teeth from both sides of the stator slot. A current level significantly above 100 mA may indicate faulty insulation or interlamination shorts at the core surface, while if a signal exceeding 300 mA this typically necessitates repair for new core.

The EL CID test on specimen generator did not identify any major anomalies with majority of reading being in the area of 100 mA, although some core segments displayed higher magnitude in the range of 250-380 mA especially at the splits and bottom of the core which were uniform across vertical span of the bore and were considered acceptable for the core of 65 years old.

F. Materials and Spare Parts

Spare bottom and top bars were available on site as well as other materials with long storage life such as wedges, side packing, ripple spring, Dacron fabric, end caps, etc. Epoxy resin components, lashing materials, and brazing equipment were supplied by OEM.

Spare bars were 30 years old and still in hermetic factory packaging. To validate the condition of the spare bars they were subjected to the series of tests to qualify as replacement specimen, as following (Table 1):

1. Insulation resistance and polarization index at 5 kV

2. Hi-Potential testing at the voltage level in DC equivalent that typically used for new winding (i.e. $1.7 \cdot (2U_{nom} + 1 \text{ kV}) \Rightarrow 48.6 \text{ kV}$)
3. Semiconductive paint of bar slot section surface resistance

TABLE I. SPARE BARS QUALIFICATION TESTING

#	Bar S/N	Insulation Resistance and Polarization Index			Hi-Potential Test		Semicon. paint
		1 min, TΩ	10min, GΩ	PI	kV DC	μA	
1	B103	325	217	6.68	51	4.8	7640
2	B105	160	219	5.37	52	6.0	7100
3	B106	132	225	4.8	51.5	4.6	7140
4	T203	175	276	6.68	50.5	3.8	8020
5	T204	175	250	6.45	50.5	2.2	6190
6	T207	236	334	6.67	52	4.0	6230

G. Winding repair

The replacement of the bar involved removing the damaged bar and installing a new one, properly insulating brazed connections, and electrically testing of new bar connections by winding resistance and winding insulation measurement [1] after repair. The new bar was installed with side ripple packing. To ensure proper installation in the slot, the contact resistance was checked (Fig. 3) between the bar surface and ground (core), with an average reading of 250 ohms. The slot wedges were replaced with a new system, including locking wedges at the top and bottom of the slot. The top ripple spring compression was checked with a gauge, measuring between 0.09 and 0.14 inches to ensure ripple spring compression consistently above 80% [7]. Diamond spacers were reinstalled in the overhang consisting of fiber-reinforced polymer (FRP G-10) wrapped with epoxy-soaked Dacron® felt (Fig.3 and 4). Connections were reinstalled as per winding diagram and insulated with wet insulation system: mica tapes and room temperature curing epoxy resin. New insulating caps were installed and filled with the capping resin at the top and bottom of the newly brazed stator bar.



Fig. 3 Bar Surface to Core Resistance Measurement and End Winding Completion

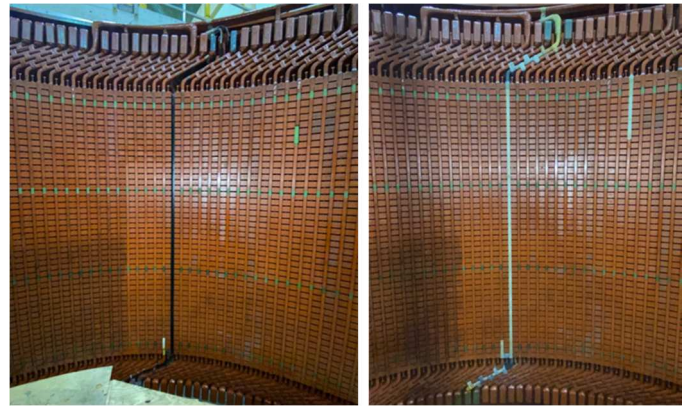


Fig. 4 Removed Bar and Replaced Bar

H. Core repair

The core area adjacent to the damaged slot was masked off with cotton cloths and duct tape to restrict the flow of liquid beyond the repair area.

The protruded loose laminations were removed from the slot tooth by coarse grinding and cutting out of the damaged iron to approximately 1.2 inch in depth. Precaution was taken to protect the stator bar from mechanical damage with the insertion of thin metal shims between the bar and the slot section.

The cutout area was treated with the process of etching until lamination separation was visible. A portable laboratory DC power supply served as the current source with the current density between 3 and 4 amperes per square inch at the contact surface. The etching was applied until the damaged laminations were seen to be distinctly separated with the naked eye. After acid etching the surface was visually inspected with a 5X magnifying glass to confirm separation of the lamination. The etching fluid was removed and a heat gun was used to dry.

The area was then treated with weeping epoxy “red eye” to seal the surface and in between of etched laminations.

The tooth replacement was reconstructed from laminate blocks of G10. The size of the block was 1.93 x 0.8” x 1.2” in height. The G10 block was large enough to form the compressed and tight fit between the finger and the lamination. The outer edge of the block was machined for the close conformation to the shape of the cavity (Fig. 5).

Room temperature cure epoxy resin was used between G10 laminates to solidify the block. After full curing of the block the wedge grooves were created for placement of the locking wedges.

Locking wedges were installed at the end to secure and hold the block in place.

The knife test was performed to check the tightness of the laminations above the repair areas. Additionally, knife check was performed in surrounding areas to ensure that they have not loosened as a result of the repairs.

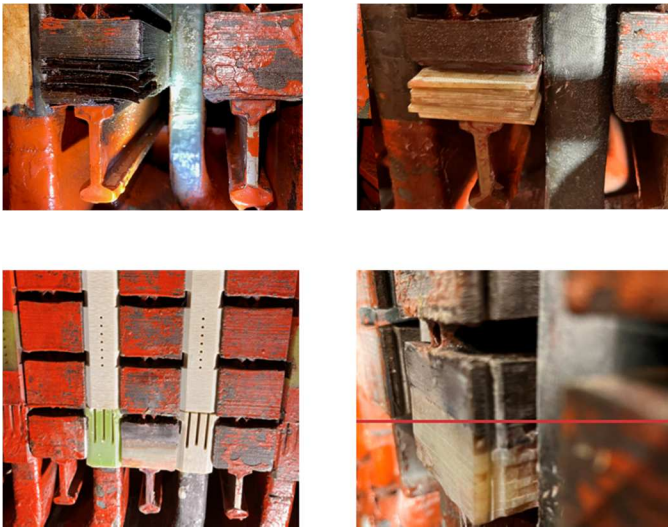


Fig. 5 Core Repair

III. ACCEPTANCE TESTING

Completed assembly was subjected to the following acceptance testing routine as per [1, 8] to ensure ready for service condition:

- Winding resistance
- Insulation resistance
- Polarization index
- HVDC ramp test
- ELCID
- Wedge tightness
- Thermocouple insulation
- Thermocouple continuity and calibration
- Validation of additional fiber optic temperature sensors to increase heat map visibility of the assembly
- Vibration signatures
- Heat run to the nominal rating

In other stator/core failure instances the set of tests can be extended to more broad selection based on the need to certify the repair, specifics of the machine design as well as operation requirements and best practices of the facility.

IV. ASSET MANAGEMENT STRATEGY

Asset management strategy considerations for stator assembly of the large generator of 100 MVA and above usually target the following aspects relative to capital and operational expenditures:

- Lifecycle;
- Load and utilization factors vs downtime cost;
- Maintenance, online monitoring and condition assessment;
- Risk management at the end of life: Repair vs Replacement Timeline

A. Lifecycle

Typical design lifespan of the stator winding is between 35-50 years and for the stator core can be as much as twice of that timeframe. However, operation at the high end of allowed temperature may significantly shorten life of stator assembly.

It is common cost-saving practice to reuse the core assembly for second rewind, however one shall consider an increase of probability to shorten winding life due to defects in the core and premature core failure in this case. If decision is made for old core to be re-used for second rewind, it is recommended to perform rigorous condition assessment and testing of the core.

As cost of new winding and its installation together with cost of the outage time is significantly (typically 5-10 times) higher than cost of new core laminations and core stacking effort it is preferable to include new core into the replacement of stator during rewind. This may allow a longer trouble free lifecycle of the asset beyond 35years.

B. Load and utilization factors vs downtime cost

For generating stations with load factor above 80% and utilization factors above 90%, connected to industrial base loads such as metal smelters and with limited connection to the bulk electric grid, reliability of the generator equipment including stator winding and core shall be above 95%.

Repair downtime costs for such facilities are very high and in some cases orderly equipment replacement is more economical option than multiple outages and subsequent repairs.

Derating of the nominal rating as a result of stator/core failure and subsequent repair can also be decisive factor to accelerate post repair replacement of the stator winding and core.

C. Maintenance, Online Monitoring, and Condition Assessment

Maintenance of the stator core assembly mainly includes cleaning winding from oil and dust residues and core ventilation ducts from obstruction, physical inspection of all electrical and mechanical components and electrical testing of winding and core.

Wedging system inspection for tightness is critical and may lead to re-wedging of the stator assembly at least once throughout the life of the stator assembly. Loose wedging system is another prime cause of the stator winding failure due to resulting excessive vibration of bar in the slot and ground wall insulation mechanical deterioration followed by electrical breakdown.

Comprehensive routine maintenance plan, procedures, and data history since commissioning tests throughout lifecycle of the stator and core play a vital role in determination of asset condition and justification of their replacement. As stator and core are considered the most expensive generator replacement subcomponents, lack of historical evidence of equipment state and deterioration usually extends the timely replacement justification of asset beyond the first end of life in service failure.

Multiple online condition monitoring tools including AI machine learning and model defined modules are available and under development for monitoring in service health of the asset and alert abnormal mode of operation. However, these tools do

not replace traditional offline inspection and diagnostics methods listed in [1, 8] that provide more conclusive results and can be used in data driven approach for magnitude of the repair and equipment life extension decisions.

D. Risk Management at The End of Life: Repair vs Replacement Timeline

Modern utilities leveraging asset management strategies to attain cost optimization, risk reduction and render operations more flexible. One of the global tasks for generation fleet owner is to develop defined road map for stator/core assembly repair and replacement strategy as a response to the increased probability of failure at the end of life.

Stator and core assembly failures account for more than 44.4% of overall generator breakdowns throughout its service [9]. Depending on the exploitation, risk of failure of the generator may significantly increase after 30-40 years of operation thus putting at risk overall power plant reliability, maintenance activities and performance metrics.

Repair tactics are a very economical way to extend machine operation and allow time to prepare for replacement. However, without stator/core engineering documentation, spare parts, repair material and procedures it may take extended amount of time to execute the quality repair. Especially important to have enough spare bars, coils, lamination, wedging and other miscellaneous materials on site at the time of machine failure to accelerate the repair process without derating of the machine. These spare parts can be included in the original stator/core supply, but also can be ordered later throughout generator life if compete engineering documentation is available, or reverse engineered.

When the repair becomes unpractical due to the extensive damage of the stator winding and/or core the replacement strategy of the components needs to be defined and ready. While direct like for like stator winding/core assembly replacement can be executed in relatively short timeframe (9-18 month), especially when reusing existing frame and core. Other alternatives such as nameplate rating uprate may take much longer time to execute (3-5 years) from project initiation and technical specification development to commissioning of the new generator.

Generator operating mode, instrumentation, maintenance practices, trained personnel and availability of the spare parts may be used to lower the risks of generator downtime at the end of life of its stator assembly.

V. CONCLUSION

In-service stator winding and core failures lead directly to generator outage and therefore impeding the operation of the power plant and its customers. Timely addressing stator winding and core failures by utilizing data driven approach for concluding a failure mode and minimizing repair consequences is crucial when load and utilization factor of the power plant are high. Comprehensive acceptance and diagnostics test methods are readily available for stator winding and core components from IEEE and other technical standards and publications and can be embedded into best maintenance, troubleshooting, and commissioning practices of the power plant or utility.

Detailed repair procedures can be available from OEM. Lack of spare parts and materials and missing engineering documentation can be addressed with the OEM by entering into master service agreements for technical support and cooperation on testing of new prototype materials, systems, and methods related to the improvement of the reliability and life extension of existing stator assemblies.

Availability of the spare parts on site and the ability to execute repair immediately upon the failure is a common task of initial stator installation project, maintenance engineering and asset management personnel. This component alone will significantly reduce a repair outage duration during stator/core assembly failure on machines above 100 MVA.

REFERENCES

- [1] IEEE Std62.2-2004-IEEE Guide for Diagnostic Field Testing of Electric Power Apparatus- Electric Machinery
- [2] IEEE Std56-2016_Guide for Insulation Maintenance of Electric Machines
- [3] IEEE C37.102-2006 - Guide for AC GEnerator Protection
- [4] IEEE Std43-2013 Recommended Practice for Testing Insulation Resistance of Electric Machinery
- [5] IEEE 95 - Recommended Practice for Insulation Testing of AC Electric Machinery (2300 V and Above) With High Direct Voltage
- [6] G.C. Stone, I. Culbert, E.A. Boulter, H. Dhirani, "Electrical insulation for rotating mashines: design, evaluation, aging, testing, and repair", 2014
- [7] I. Chichkin, "Stator winding wedge tightness evaluation on large hydro generators", 2022 IRIS Rotating machines conference.
- [8] IEEE Std 115-2019, IEEE Guide for Test Procedures for Synchronous Machines Including Acceptance and Performance Testing and Parameter Determination for Dynamic Analysis
- [9] O. Savin, C. Badina, J.L. Drommi, J. Baroth, S. Charbonnier, C. Berenguer. Influence of Starts and Stops on the Aging of Hydroelectric Generator Stators by Thermal Cycling: Empirical Study and Accelerated Lifetime Model. ESREL 2021 - 31st European Safety and Reliability Conference, Sep 2021, Angers, France. pp.3214-3221, 10.3850/978-981-18-2016-8_315-cd.hal-03443019